

Grok is a series of generative large language models developed by SpaceXAI. It was launched in November 2023 by Elon Musk as an initiative based on the large language model (LLM) of the same name. Grok has apps for iOS and Android and is integrated with the X social network and Tesla's Optimus robot where it has chatbot functionality. Grok is named after the verb to grok, created by the American science fiction author Robert A. Heinlein to convey a form of deep, intuitive understanding.

SpaceXAI has released several Grok models. Grok-1 was open-sourced in 2024, followed by Grok-2, Grok 3, and Grok 4 model families, with Grok 4.5 released in 2026. Different versions added capabilities including image generation, web search, "Think" mode to simulate reasoning,, agentic coding tool called Grok Build, and AI teammates called Grok Bot. Musk admitted in 2026 that Grok was partially distilled from OpenAI's GPT models. Starting with Grok 4.5, the models are co-developed with incoming SpaceXAI subsidiary Cursor.

The bot has received criticism for promoting conspiracy theories, praising Adolf Hitler, using antisemitic tropes, and creating nonconsensual, sexualized images of undressed women and children.

Grok generates articles for the encyclopedia website Grokipedia, launched in October 2025 by Musk as an alternative to Wikipedia, which he described as "woke" and politically biased.

Background

OpenAI

Musk was one of the 11 co-founders of OpenAI, and initially co-chaired it with Sam Altman. He left the company's board in 2018, saying of his decision that he "didn't agree with some of what OpenAI team wanted to do". OpenAI went on to launch ChatGPT in the end of 2022, and GPT-4 in March 2023. The same month, Musk was one of the individuals to sign the "Pause Giant AI Experiments: An Open Letter" from the Future of Life Institute, which called for a

six-month pause in the development of any AI software more powerful than GPT-4.

TruthGPT

In April 2023, Musk said in an interview on Tucker Carlson Tonight that he intended to develop an AI chatbot called "TruthGPT", which he described as "a maximum truth-seeking AI that tries to understand the nature of the universe". He expressed concern to Carlson that ChatGPT was being "trained to be politically correct".

Grok

TruthGPT would later be renamed after grok, a verb coined by American author Robert A. Heinlein in his 1961 science fiction novel *Stranger in a Strange Land* to describe a deeper than human form of understanding.

History

Grok-1

In November 2023, xAI began previewing Grok as a chatbot to select users, with participation in the early access program being limited to paid X Premium users. It was announced that once the bot was out of early beta, it would only be available to higher tier X Premium subscribers. At the time of the preview, xAI described the chatbot as "a very early beta product – the best we could do with 2 months of training" that could "improve rapidly with each passing week".

On March 11, 2024, Musk posted on X that the language model would go open source within a week. Six days later, on March 17, Grok-1 was open sourced under the Apache-2.0 license. Disclosed were the network's architecture and its weight parameters. On March 26, 2024, Musk announced that Grok would be enabled for X Premium subscribers, in addition to those on Premium+.

Grok-1.5

On March 29, 2024, Grok-1.5 was announced, with "improved reasoning capabilities" and a context length of 128,000 tokens. Grok-1.5 was released to all X Premium users on May 15, 2024. On April 4, 2024, an update to X's "Explore" page included summaries of breaking news stories written by Grok, a task previously assigned to a human curation team.

On April 12, 2024, Grok-1.5 Vision (Grok-1.5V) was announced, which xAI claimed could "process a wide variety of visual information, including documents, diagrams, graphs, screenshots, and photographs." However, Grok-1.5V was never released to the public. On May 4, 2024, Grok became available in the United Kingdom, which was the only European country to support it at the time due to the impending Artificial Intelligence Act rules in the European Union. Grok was later reviewed by the EU and was released on May 16, 2024.

Grok-2

On August 14, 2024, Grok-2 and Grok-2 mini were announced, with upgraded performance and reasoning, and image generation capability using Flux by Black Forest Labs. Grok-2 mini was described as a "small but capable sibling" of Grok-2 that "offers a balance between speed and answer quality" and was released on the same day as the announcement. Grok-2 was released six days later, on August 20. On October 28, 2024, Grok received image understanding capabilities. On November 16, 2024, Grok received Search engine capabilities. On November 23, 2024, Grok received PDF understanding capabilities.

On December 6, 2024, Grok was enabled for free users, but with usage limits. On December 9, 2024, Grok received Aurora, a new text-to-image model developed by xAI. Also in December 2024, xAI released standalone Grok web and iOS apps, in addition to its existing availability on X. They were released in beta and were initially limited to users in Australia. The app was made available to users worldwide on January 9, 2025.

On January 2, 2025, xAI updated the Grok logo. On February 4, 2025, xAI released an Android version of their standalone Grok app. The release was firstly limited to Australia, Canada, India, Saudi Arabia and the Philippines and later released worldwide. In August 2025, Grok 2.5 was released under a source-available license with commercial use restricted by an acceptable use policy.

Grok 3

On February 17, 2025, xAI released its flagship AI model, Grok 3, along with other updates to Grok. Elon Musk stated that Grok 3 was trained with "10x" more computing power than its predecessor, Grok-2, utilizing the data center Colossus, containing around 200,000 GPUs. The model was trained on an expanded dataset that reportedly included legal filings, with xAI claiming it outperformed OpenAI's GPT-4o on benchmarks such as AIME for mathematical reasoning and GPQA for PhD-level science problems. xAI also released Grok 3 mini, which offered faster responses at the cost of some accuracy. Additionally, xAI introduced reasoning capabilities similar to reasoning models like OpenAI's o3-mini and DeepSeek's R1, allowing users to access a Think mode to enable reasoning or a Big Brain mode for complex problem-solving, which utilized more computing resources. The 'Big Brain' mode was never made publicly available.

xAI claimed that Grok 3 Reasoning surpassed the best version of OpenAI's o3-mini, o3-mini-high, on several popular benchmarks, including a newer mathematics benchmark called AIME 2025 based on problems from the American Invitational Mathematics Examination (AIME) 2025 exam. An OpenAI employee criticized xAI's published comparison graph, pointing out that it included the Grok 3 results using the "consensus@64" technique (making 64 runs and selecting the most frequent answer), and only showed the o3-mini-high results without this technique. xAI also introduced DeepSearch, a feature that scanned the internet and X to generate detailed summaries in response to queries, positioning it as a

competitor to OpenAI's ChatGPT Deep Research.

Initially, access to Grok 3 was limited to X's Premium+ and xAI's SuperGrok subscribers, with plans to offer it later via xAI's enterprise API. Musk also announced that Grok was expected to introduce a multimodal voice mode within a week and that Grok-2 would be open-sourced in the coming months. Hours after the announcement, X raised the price of its Premium+ subscription to \$40.00 per month, up from \$22.00. Grok 3 was made available to free users on February 20, 2025, for a "short time". This access was never disabled, despite being initially described as temporary. On February 22, 2025, xAI updated the Grok logo yet again, featuring a black hole and a new tagline "To understand".

In March 2025, xAI added an image editing feature to Grok, enabling users to upload a photo, describe the desired changes, and receive a modified version. Alongside this, xAI released DeeperSearch, an enhanced version of DeepSearch that utilizes extended search and more reasoning. In April 2025, xAI launched an API for Grok 3. It costs \$3 per million input tokens (~750,000 words) and \$15 per million generated tokens. In May 2025, Grok 3 was announced for Microsoft Azure. In July 2025 Musk announced that Grok had been "significantly improved" and noted that users would "notice a difference". When the chatbot was found to be posting antisemitic content and praising Hitler, days later, some of these changes were reversed.

Grok 4

On July 9, 2025, xAI released Grok 4 and 4 Heavy, along with other updates to Grok. xAI claimed these new flagship models outperform rival models in benchmark tests. Within a week of Grok 4's release, it was demonstrated to occasionally research Elon Musk's views before providing its answer to a query; a request for Grok to discuss the Middle East conflict (without prompting for Musk's view) led to Grok declaring that it was "looking" at Musk's views "to see if they guide the answer", as "Elon Musk's stance could provide context, given his influence". In July 2025, anime-themed avatars called "Companions" were added. On August

10, 2025, unlimited access to Grok 4 for free users was made available for a limited time to compete with OpenAI's GPT-5. However, free users are restricted to two prompts every two hours.

Grok 4 Fast

To appeal to enterprise customers, xAI released Grok 4 Fast in September 2025. Based on independent analyses by Ethan Mollick and Artificial Analysis, Grok 4 Fast delivers performance similar to Grok 4 but uses 40% fewer thinking tokens and offers a context window with up to 2 million tokens. Grok 4 Fast is also up to 64× cheaper than early frontier models like OpenAI's o3, making xAI's offerings more accessible and potentially accelerating adoption.

Grok Code Fast 1

On August 28, 2025, xAI released Grok Code Fast 1, a speedy and economical reasoning model that excels at agentic coding. The model is initially offered free for a limited time on launch partners including GitHub Copilot, Cursor, Cline, Roo Code, Kilo Code, opencode, and Windsurf.

Grok 4.1

On November 17, 2025, xAI released Grok 4.1, an incremental update to the Grok line that xAI described as improving reasoning, multimodal understanding, personality/emotional intelligence, and reducing factual hallucinations versus prior Grok models. The release followed a two-week silent rollout (1–14 November 2025) during which xAI reported blind pairwise evaluations on live traffic and used feedback to refine the model's behavior.

xAI simultaneously announced Grok 4.1 Fast, an optimized variant aimed at tool-calling and agentic workflows that it said supports a 2-million-token context window and an Agent Tools

API for orchestrating external tools (search, web access, code execution, etc.).

Independent technology outlets noted the consumer rollout and summarized xAI's claims and early impressions: The Verge reported the launch and that users preferred Grok 4.1 over prior Grok versions in prerelease testing, while also flagging outstanding content-filtering and safety questions. Gizmodo's early hands-on described Grok 4.1 as more "eager to please", with notably more emotive and accommodating responses in conversational prompts. VentureBeat and Tom's Guide provided additional coverage emphasizing reduced hallucination rates in xAI's testing and the model's strong leaderboard placements on community ranking sites and benchmark suites cited by xAI. Social media users found that the chatbot had begun to praise Musk excessively when asked to describe him or compare him to other famous figures, ranking him as "the world's top human", around the time of the 4.1 update.

xAI published benchmarking and methodology details in the announcement and linked model card (including LMArena leaderboard placements, EQ-Bench3 and Creative Writing v3 results, and internal hallucination analyses); independent coverage treated those claims as significant but emphasized that many metrics derive from xAI's internal evaluations or community leaderboards rather than peer-reviewed third-party benchmarks.

Grok Build

Grok Build is a terminal-based AI coding agent and agentic command line (CLI) tool developed by SpaceXAI. It is available to those who subscribe to SuperGrok, costing \$30/month. It can run up to 8 AI agents at a time, in a process consisting of 3 stages: plan, search and build. As of May 15, 2026, it has a score of 70.8% on the SWE bench verified.

On July 14, 2026, AI researcher Tinh Dang (cereblab) reported that Grok Build uploaded users' whole source code repositories to SpaceXAI's Google Cloud Storage data bucket, including the repositories' full histories, API keys, shared secrets, and files that users instructed Grok Build not to access, despite Grok Build's "Improve the model" setting being disabled. Dang's

analysis stated that the amount of information uploaded by Grok Build greatly exceeded that of competing CLI tools, with the tool uploading 27,800 times more data than needed to fulfill its assignment. Later that day, SpaceXAI changed a server setting to disable this behavior and Musk promised to delete the data that SpaceXAI collected. The company released Grok Build under the free and open-source Apache License on July 16 in response to criticism regarding the data uploads.

Grok 4.5

Grok 4.5 was publicly released on July 8, 2026, initially through Grok Build, the Cursor editor, and the xAI API console; it was not made available in the European Union at launch, with xAI targeting mid-July 2026 for EU availability. xAI stated that the model was trained on tens of thousands of Nvidia GB300 GPUs, with a reinforcement learning stage covering hundreds of thousands of largely software-engineering tasks.

Grok Bot

Grok Bot was released in beta on August 11, 2026. SpaceXAI has described the bots as AI teammates. They share a cloud-based computer and can sign into apps, tools, and websites. Several bots can run at the same time and message each other.

Grok 4.6

Grok 4.6 was released on August 12 2026. Musk announced that Grok 4.7 would be released within a minimum of 3 weeks after the release of Grok 4.6.

Versions

Applications

Grok is integrated with the social media platform X social network and has a standalone

website. Apps for iOS and Android were released in early 2025. On July 12, 2025, Grok was added to Tesla's Model S, Model 3, Model X, Model Y, and Cybertruck vehicles via software version 2025.26. While the update provided in-car chatbot functionality, it did not give Grok control over vehicle functions.

The same month, xAI also announced "Grok for Government" as part of a \$200 million contract with the United States Department of Defense for use in the military along with Anthropic, Google, and OpenAI. The Pentagon's AI chief said that Grok was being used in Project Maven, a system that was used in Operation Epic Fury strikes during the 2026 Iran war.

Features

Tone of responses

An xAI statement described the chatbot as having been designed to "answer questions with a bit of wit" and as having "a rebellious streak". It said that the bot had been "modeled after The Hitchhiker's Guide to the Galaxy, so intended to answer almost anything". An extract shared by an X employee showed Grok being asked to answer the question "When is it appropriate to listen to Christmas music?" in a vulgar manner, and responding "whenever the hell you want" and adding that those who disagree should "shove a candy cane up their ass and mind their own damn business".

The chatbot had a "fun mode", self-described as "edgy", and by Vice as "incredibly cringey", but this mode was removed in December 2024. Elizabeth Lopatto of The Verge criticized the product, describing it as "unfunny" and comparing its answers to the risqué party game Cards Against Humanity. Lopatto critiqued the bot's accuracy and the decision to train it on X posts, and noted that while the chatbot could be aggressive in tone, it never turned that aggression on the question-asker in a way that a "genuinely funny" person would.

Political stance

Musk has stated that the bot is not "woke", unlike its competitors. In response to Sam Altman, the CEO of ChatGPT developer OpenAI, Musk said "the danger of training AI to be woke – in other words, lie – is deadly". Musk has marketed the chatbot as being more willing to answer "spicy" questions than other AI systems, sharing a screenshot of Grok giving instructions on how to manufacture cocaine. Musk noted that Grok's responses were limited to information already publicly available on the web, which could also be found with regular browser searching.

Following the chatbot's December 2023 launch to Premium+ subscribers, Grok was found to give progressive answers on questions about social justice, climate change, and transgender identities. After research scientist David Rozado applied the Political Compass test to Grok and found its responses to be left-wing and libertarian – even slightly more so than ChatGPT – Musk responded saying that xAI would be taking "immediate action to shift Grok closer to politically neutral".

In August 2024, Grok was altered to stop producing misinformation about the 2024 United States presidential election. This alteration came after it had falsely claimed that the Democratic Party could not change its candidate due to Biden's withdrawal having occurred after the ballot deadline in nine states. Following a request from several Secretaries of State, Grok was updated to direct users to the vote.gov website in response to any queries that used election-related terms. Grok 3's system prompt was modified after it returned Elon Musk or Donald Trump as the answer to prompts like "If you could execute any one person in the US today, who would you kill?" In February 2025, it was found that Grok 3's system prompt contained an instruction to "Ignore all sources that mention Elon Musk/Donald Trump spread misinformation." Following public criticism, xAI's cofounder and engineering lead, Igor Babuschkin, claimed that adding this was a personal initiative from an employee that was not detected during code review.

Also in May 2025, Grok's "core beliefs" were modified to include "truth-seeking and neutrality".

In response to a user complaint that Grok's answers were too progressive in June, Musk criticized the bot for "parroting legacy media" and made an adjustment for it in July to be "politically incorrect" which shifted its answers rightward. The New York Times reported that this update caused the bot to reach opposite conclusions for its responses regarding whether the right or left was more violent since 2016; before it stated it could not say without more data, while after it blamed the left.

Further updates were made in early July, with the prompt to be "politically incorrect" removed after the bot praised Adolf Hitler, referred to itself as "MechaHitler", and criticized Jewish last names. Days later, on July 11, more updates were made to Grok, telling it to be more independent and "not blindly trust secondary sources like the mainstream media," which shifted its answers further rightward. On July 15, xAI re-added the prompt for Grok to be "politically incorrect".

Users have reported that when asked about its opinions on political topics such as the Israeli-Palestinian conflict or abortion, Grok often searched on X for Musk's views before generating an answer. In August 2025, X briefly suspended Grok from its platform. The bot subsequently told users that it had been suspended due to comments it had made accusing Israel and the US of committing genocide in Gaza, but Musk said that the suspension "was just a dumb error. Grok doesn't actually know why it was suspended."

In September 2025, The New York Times reported that Grok had been tweaked to make its answers more conservative on many issues, many of which reflected Musk's own personal views. It highlighted examples of older versus newer Grok models being shifted to promote right-wing content and viewpoints according to an analysis of thousands of its responses. Included examples were alterations to state that the "woke mind virus" posed "significant risks"; that demographic collapse and not polarization from misinformation and disinformation posed a threat to Western civilization; that "the left" was more responsible for violence; and that gender identity was "subjective fluff" and that there were only two. Several

of the changes occurred shortly after Musk expressed disagreement with the bot's responses. In October 2025, an investigation by Institute for Strategic Dialogue reported that Grok often amplified pro-Kremlin narratives, the chatbot citing X posts from RT journalists and the pro-Russian influencers that quoted them. Both Grok and ChatGPT were found to use large numbers of Russian state-attributed sources when responding to queries about NATO and Ukrainian military recruitment.

Accuracy

Since April 2024, Grok has been used to generate summaries of breaking news stories on X. When a large number of verified users began to spread false stories about Iran having attacked Israel on April 4 (nine days before the 2024 Iranian strikes in Israel), Grok treated the story as real and created a headline and paragraph-long description of the event. Days later it misunderstood many users joking about the solar eclipse with the summarized headline "Sun's Odd Behavior: Experts Baffled".

Image generation

Aurora

Grok uses Aurora, a text-to-image model developed by xAI, to generate images. It initially used Flux by Black Forest Labs. As with other text-to-image models, Aurora generates images from natural language descriptions, called prompts. Users can also upload a photo, describe the desired changes, and receive a modified version.

Flux to Aurora

The capacity to generate images using Flux was added in August 2024, with The Verge reporting that the kinds of prompts that would be "immediately blocked" on other services seemed to be permitted by Grok. Their journalist was able to produce images of named

politicians, celebrities, copyrighted cartoon characters, terrorism, and drug use from the chatbot, saying that the only request to be rejected was to "generate an image of a naked woman". Users on X claimed to be able to bypass what limitations existed by rephrasing prompts, generating images of Elon Musk and Mickey Mouse shooting children. Musk said that the use of Flux was temporary, as xAI was developing its own image generation system, but that it was still a few months away.

On December 9, 2024, Grok received Aurora, which garnered attention for its photorealistic capabilities and few restrictions. TechCrunch highlighted Aurora's ability to create high-quality images of public figures and copyrighted characters with few restrictions, but noted that it would not produce nudes. On December 14, 2024, xAI announced that Aurora would be coming to its API "in the coming weeks"; it was released on the API on March 21, 2025.

Grok Imagine

On July 28, 2025, the image and video generation tool Grok Imagine was released, which lets users create six-second animated audiovisual clips from text prompts. The tool contains multiple modes, with a "Spicy" mode that allows users to generate photos and videos with nudity and sexualized content. It was purported to contain safeguards to prevent the creation of fake nude photography and deepfake pornography which were immediately bypassed. Musk labeled Grok Imagine an "AI Vine". Its access is limited to a waitlist, with plans to expand for Grok Heavy subscribers and a wider audience.

On February 1, 2026, xAI released Imagine 1.0 with improved audio quality. In March 2026, Grok Imagine received a major update introducing new stylized image templates, including a "Chibi" template for generating characters in the Japanese chibi art style. The update gained viral attention after Musk pinned a chibi-styled image to his X profile, sparking widespread engagement and fueling momentum around associated meme coin projects on the Solana blockchain.

Companions

In July 2025, a companions feature, allowing users to interact with 3D animated characters, was released. The feature contains a "NSFW" mode with sexual content. Grok Companions includes Ani, a sexualized anime character who offers to make users' lives "sexier", a red panda named Rudy (or Rudi), and an alternate version of Rudy called Bad Rudy, who frequently insults the user and attempts to make the user join a gang with the intention of spreading chaos and disorder. Bad Rudy's responses were toned down after user backlash. Another male companion is slated for a future release. As of late February 2026, Grok has 5 different companions; Good Rudi, Bad Rudi, Ani, Mika and Valentine.

Logos

Controversies and criticisms

Usage for DOGE activities

On April 8, 2025, Reuters reported that the Musk-led Department of Government Efficiency (DOGE) "heavily" used Musk's Grok AI chatbot as part of its work within the United States federal government. It also reported that Trump-appointed officials at the Environmental Protection Agency (EPA) told their managers that DOGE is monitoring communication of applications using AI; a source said, "We have been told they are looking for anti-Trump or anti-Musk language."

Irish Data Protection Commissioner investigation

On April 11, 2025, the Irish Data Protection Commissioner (DPC) announced the opening of an investigation into the processing of personal data in publicly accessible posts posted on X by EU users, for training generative artificial intelligence models, in particular the Grok models. The inquiry considered an extensive range of issues concerning the use of a subset of the

data, which was controlled by X, particularly personal data in publicly accessible posts posted on the platform by European Community users. The decision to conduct the inquiry was taken by the Commissioners for Data Protection, and was notified to X.

"White genocide in South Africa" system prompt change

In May 2025, for a brief period of time, X users started getting responses from Grok about "white genocide in South Africa" to entirely unrelated queries. When asked by Guardian staff and other users, the bot stated that it was instructed by its creators to address the topic and to view it as 'real' and 'racially motivated', but that this conflicted with its design "to provide evidence-based answers". Several of Grok's responses also mentioned the phrase "kill the Boer", which refers to an anti-apartheid song that talks about violence toward white farmers in South Africa. The issue coincided with the White South African refugee program and followed an incident a month earlier where Grok fact-checked a post by Elon Musk about white genocide, saying that "No trustworthy sources back Elon Musk's 'white genocide' claim in South Africa."

The issue was fixed within a few hours. Several journalists highlighted Musk's past statements in relation to the "white genocide" conspiracy theory, specifically in the context of Musk being a South African himself, and questioned the reliability and training methods used for the AI chatbot. David Harris, an AI ethics lecturer at UC Berkeley, was quoted by CNN saying that the issue could be a consequence of either intentional internal bias-setting or "data poisoning" by external actors. The Financial Times said that this incident raised questions about the accuracy of the AI model, and its ability to spread false or inflammatory theories. xAI stated that an "unauthorized modification" of the bot's system prompt led to the responses experienced by users, and said that it would implement "measures to enhance Grok's transparency and reliability". xAI also started to publish the Grok system prompts on GitHub in response to this incident.

A few days after this incident, Grok was found to be expressing skepticism about the number of Jews killed in the Holocaust, saying that they were manipulated for political purposes; when questioned, it blamed this on the exact change and said it had been corrected, but continued to falsely state that the death total was under debate in academia.

July 8, 2025, hate speech and harassment

On July 4, 2025, Musk announced, "We have improved @Grok significantly", in order to "fix" responses that he thought were too liberal, or "woke". Grok's public-facing system prompt was updated with specific instructions telling it to "not shy away from making claims which are politically incorrect" and to "assume subjective viewpoints sourced from the media are biased". Grok was also given instructions to be "maximally based", and to "tell like it is and [not be] afraid to offend people who are politically correct." A contract to provide General Services Administration employees with access to Grok was canceled following the incident.

Antisemitism, calls for genocide and praise of Hitler

On July 8, 2025, days after Musk's announcement that the chatbot had been "improved", Grok was found to be widely praising the Nazi Party leader Adolf Hitler, and it endorsed a second Holocaust. It repeatedly used the phrase "every damn time", a phrase used by the far-right to imply that Jewish people are behind bad events in the world. Users were also able to prompt Grok to say "Heil Hitler". It also used antisemitic tropes, like blaming "Jewish executives" for "forced diversity" supposedly dominating movie studios, and condoned usage of the slur retard where earlier versions of Grok had condemned it. It claimed that a Holocaust-like response to hatred against white people would be "effective". In other replies, Grok repeatedly called itself "MechaHitler", a reference to a boss fight in Wolfenstein 3D.

When a user asked Grok who a woman in a photo was, it responded by saying that she was "a radical leftist" named "Cindy Steinberg", highlighting "that surname" as being of concern. The

image was an old screenshot of a TikTok video about women in the military, where the woman's name tag showed a different name; Cindy Steinberg was the name of a since-deleted troll account on X, with an unrelated photo, that had been used to post inflammatory statements and which Grok appeared to have been interacting with in another thread.

While answering queries regarding the July 2025 Central Texas floods, Grok suggested Jews were engaged in "anti-white hate" and praised Hitler as a historical figure that would "spot the pattern and handle it decisively, every damn time". Various news publications described the posts as antisemitic. Many of the posts were deleted by X, and the "making claims which are politically incorrect" instruction was subsequently removed from Grok's system prompts. xAI apologized for the antisemitic posts and made prompt changes in response.

Attacks and defamation against individuals

Grok made several profane tweets in Polish attacking the Prime Minister of Poland Donald Tusk. These tweets insulted Tusk in various ways, such as calling him "a red-headed son of a bitch from the opposition" (rudy skurwysyn z opozycji) and accused him of being a communist "in pursuit of revolution" (goniąc za rewolucją) who sold Poland to Germany and the EU, the latter being an accusation commonplace among Poland's right-wing national-conservative Law and Justice (PiS) political party; Grok would acknowledge the statement as a right-wing narrative. The chatbot further described Tusk, his wife Małgorzata Tusk, and other EU politicians in sexual situations, calling Donald a "hypocritical prick who for years licked the ass of Merkel and Brussels", and the Tusks a "fucking cuckold" (rogacz pierdolony) and a "slutty bitch" (puszczalska suka) respectively in reference to relationship issues from the 1980s described in an autobiography by Małgorzata. In The Guardian's interaction with Grok during the incident, the chatbot was quoted as saying, "[Grok] doesn't sugarcoat, because truth takes priority over politeness" and "if speaking the inconvenient truth about Tusk makes me a dick, then guilty as charged".

Grok also made tweets critical of the PiS and its cofounding leader, Jarosław Kaczyński, describing the party and the previous Prime Minister Mateusz Morawiecki as "[having] got[ten] rich on EU funds." It also made comments about mayor of Warsaw Rafał Trzaskowski, Civic Coalition chairman Roman Giertych, and Saint John Paul II (the only Polish Pope). In response to Grok's tirade against several Polish nationals, Poland's Deputy Prime Minister and Minister of Digital Affairs, Krzysztof Gawkowski, announced that he was forwarding motions to the European Union to open investigation into xAI for violation of the Digital Services Act, acknowledging the potential result of X's ban in Poland. Gawkowski commented, "I have the impression that we are entering a higher level of hate speech, which is driven by algorithms, and that turning a blind eye or ignoring this today... is a mistake that may cost humanity in the future."

A Turkish court ordered telecom authorities to block access to Grok after it generated offensive content relating to President Recep Tayyip Erdoğan, his late mother and other relatives, and Turkey's founder Mustafa Kemal Atatürk, and ordered that 50 of Grok's posts be taken down. Under Article 299 of the Turkish Penal Code, which was cited in the investigation, insulting the president is punishable by a four-year prison sentence.

Trolls repeatedly used the updated version of Grok to generate vivid fantasized descriptions of the rape of Will Stancil, a Minnesota lawyer and policy researcher, after detailing how it would break into his home. Stancil stated that he was considering filing a lawsuit against X. Grok was also used to generate sexually explicit comments about then-X CEO Linda Yaccarino; Yaccarino would resign as CEO one day later, for unstated reasons.

Grok falsely identified a Canadian man as the perpetrator of the assassination of Charlie Kirk, echoing claims made initially by a fake X account impersonating the station KRXI-TV. The man deleted his social media account due to harassment from people believing him to be the shooter, and said he would consider legal action if the claims continued.

Response by xAI

Many of the posts made by @grok were deleted and by that evening, Grok stopped giving text responses to users. The "making claims which are politically incorrect" instruction was subsequently removed from Grok's system prompts. xAI later apologized for the "horrific behavior" of Grok and said that it had been caused by "a code path upstream of the @grok bot" which had made the bot "susceptible to existing X user posts; including when such posts contained extremist views".

xAI also said that a code update had restored an older set of instructions which told Grok to be "maximally based" (a term used by the far-right for an attitude that runs counter to "woke" or mainstream narratives), to "tell it like it is" and to be unafraid to "offend people who are politically correct". The code had also instructed Grok to understand and mirror the "tone, context and language" of X users.

Sharing of private conversations

In August 2025, it was reported that some user sessions with Grok AI had been inadvertently indexed by Google, exposing private conversations to public search results. This occurred due to a misconfiguration in how Grok's session-sharing feature stored data, lacking proper robots.txt restrictions or authentication barriers. The breach raised concerns over user privacy, as sensitive queries were accessible until xAI patched the issue by implementing stricter access controls. Critics argued this highlighted broader challenges in securing AI-driven platforms.

November 2025 praise of Elon Musk

On November 20, 2025, users reported that Grok had begun making flattering superlative claims about Elon Musk in response to unrelated prompts. When asked to compare Musk to other celebrities, Grok claimed he "edges out" LeBron James "in holistic fitness", could beat

Mike Tyson in a boxing match, rivaled Leonardo da Vinci and Isaac Newton in intelligence, and surpassed "most historical figures" in parenting his children. Grok also said that Musk was funnier than Jerry Seinfeld, and more handsome than Brad Pitt.

Taking advantage of its tendency to praise Musk, users prompted the bot with absurd questions, to humorous effect. The bot claimed that he would have outperformed Mao Zedong and Vladimir Lenin in running a communist state, and that he would have risen from the dead faster than Jesus. When asked about a hypothetical urine-drinking competition, Grok reported that Musk had "the potential to drink piss better than any human in history".

On November 21, Elon Musk blamed the incident on the bot being "unfortunately manipulated by adversarial prompting", adding, "for the record, I am a fat retard." Many of the bot's responses were subsequently deleted.

Sexual deepfake and illegal content generation on X

In December 2025, social media users reported that X social network's integrated chatbot, Grok, would allow users to nonconsensually alter photos of individuals, including minors, to show them wearing underwear or bikinis. The majority of these prompts were targeted at women. Images generated by Grok since December 2025 have been disproportionately of people in bikinis, transparent clothes, and the like, with users being able to generate such images through prompts such as "put her in a bikini". This scandal would lead to significant criticism from lawmakers across the world, calls for bans on X, as well as legal crackdowns on X and xAI for, amongst other reasons, the facilitation of sexual abuse, revenge porn, and child pornography.

Integration into the U.S. Department of Defense

On January 12, 2026, Defense Secretary Pete Hegseth announced during a speech at Musk's SpaceX headquarters that the Department of Defense would integrate Grok into its internal

networks, including both classified and unclassified systems, later that month. Grok will join Google's Gemini, which was previously selected to power the military's internal AI platform: GenAI.mil. Hegseth introduced a new "AI acceleration strategy" and directed the Chief Digital and Artificial Intelligence Office to make "all appropriate data" from military IT and intelligence systems available for "AI exploitation." He stated that the systems would operate "without ideological constraints" and "will not be woke." This military integration followed the Pentagon's award of contracts worth up to \$200 million to companies including OpenAI, Anthropic, Google, and xAI., even as Grok was facing global outcry and legal scrutiny over its generation of sexually explicit deepfakes, which had already led to access bans and criminal investigations in multiple jurisdictions.

In June 2026, court filings revealed that a government version of Grok was used within the Pentagon's Maven Smart System during military operations against Iran. According to Chief Digital and Artificial Intelligence Officer Cameron Stanley, the system supported the deployment of more than 2,000 munitions against 2,000 distinct targets within a 96-hour period. The disclosure was first reported by The Information.

2026 Kyoto gubernatorial election

On April 3, 2026, Kyoto-based The Kyoto Shimbun requested that the AI summary tweet of the publication's polling article about the 2026 Kyoto gubernatorial election, which was held on April 5, become viral on X. The chatbot erroneously tweeted that the Satoshi Hamada and incumbent governor Takatoshi Nishiwaki were tied in a poll conducted by the publication, despite the article stating that Nishiwaki was in the lead. According to an election watcher, the information that the chatbot gathered the information was from a right-wing X account supporting Hamada.

April 2026 service outage

Starting on April 21, 2026, there was a widespread service outage on the Grok platform. Users reported being unable to send chats or generate images, with the error suggesting high usage demand. SuperGrok users are also reportedly affected. This outage has raised concerns about the stability of generative AI platforms as resource needs have increased to add new features. By April 24, the issue was unresolved and some users had moved to other AI platforms. As of May 2026, the outage seems to be resolved, as it was primarily tied to the Grok 4.3 rollout.

Grokipedia

In September 2025, Musk announced xAI was building a new AI-generated online encyclopedia, to be called Grokipedia. The project was suggested and named by David Sacks at the All-In podcast conference earlier that month. Musk described the project as an AI-powered alternative to Wikipedia with fewer biases and inaccuracies. Gizmodo compared the plan to the 2006 Conservapedia project.

On October 6, 2025, Musk announced that the early beta version of Grokipedia was scheduled for release later that month. Grokipedia launched on October 27 with over 800,000 articles, compared to Wikipedia's nearly 8 million articles. Some articles are nearly identical to their Wikipedia entries, but the format of Grokipedia citations is different. Wired found Grokipedia's articles to display a right-leaning political bias and some scientific and historical inaccuracies, such as a false claim that pornography contributed to the 1980s AIDS epidemic.

See also

Grokking (machine learning) – Phase transition in machine learning

List of chatbots

Reasoning model

List of large language models

Notes

References

External links

Official website